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StockGAP: Generalized Article Predictor for Stock Prices from Category-Based News
Project Final Report
(GitHub link: <https://github.com/AlexI-J/EECS-487-Final-Project>)

1. Problem Description

News articles are a frequent method by which stock traders obtain insights, allowing them to make valuable investments in certain stocks. However, beyond the traditional approach of reading financial articles, more general news can also provide important context behind stock changes. This poses a challenge when constructing detailed trends for individual stocks over time. In this project, we will train a large language model to identify how news articles impact a select list of stocks in the stock market. Specifically, we will answer the question: Does including all categories of news provide a good context for predicting changes in stock prices? By allowing accurate stock predictions to be made from a broader scope of news articles, we hope to make the research and planning that goes into stock trading more accessible to a wider audience.

Our task is to make a prediction of whether prices for a series of popular stocks will increase or decrease on average. To accomplish this, we have developed and released a web application written in Python, which uses the FastAPI and Streamlit libraries to allow the end user to train and test models easily. First, the user is prompted to train a model on stock and news data from a window of time they choose, anywhere within the range January 1, 2016 to April 1, 2020. To use the trained model on new data, the user enters a list of headlines from news articles. They will receive one of three possible predictions: the average stock price for our selected stocks will be predicted to either increase, decrease, or remain the same.

The contributions of each of our team members to the project were as follows: Andre developed the methodology for preprocessing the data into CSV files and the Streamlit frontend accessible by the end user, Alex collated this data further into a Python class that works for training a machine learning model, and Winson designed an FFNN model in the sklearn library to accomplish our goal. All team members actively participated in the development of our application, including specifying what problem we wanted to solve and implementing a solution using our source data.

2. Reference/Related Work

Various research projects have combined financial news data with sentiment analysis to effectively train NLP models, such as the Financial News and Stock Price Integration Dataset or

FNSPID (Dong et al.), as well as FinBERT-LSTM (Ma et al.). Additionally, a DRNews model has been developed to use dense vector representations of news articles to facilitate event extraction (Gu et al.). However, these projects are focused on strictly financial articles, and they undervalue the importance that current events as a whole have in influencing trends in the stock market. By contrast, our project compares the effectiveness of non-financial articles and financial articles in predicting trends. Additionally, rather than using a pretrained model like the Newsflow project, we have trained our own LLM from scratch for this purpose.

3. Methodology

3.1. Data Preprocessing

We used a combination of two datasets for this project: one source of generalized news articles, and one source of numerical, historical stock prices.

3.1.1. News Data

The news articles used to train the model were sourced from the All the News 2.0 Dataset by Andrew Thompson, which features over 2.6 million news articles from 27 publications in the United States, with a time frame from January 1, 2016 to April 2, 2020. The download link to this dataset contains a zipped archive with the CSV file “all-the-news-2-1.csv”, which is very large when uncompressed, at a file size of 8.16 GB.

Our interest in this dataset focused on the content of the article text. Therefore, the following columns from the original dataset were dropped:

- **year, month, day:** The year, month, and day (respectively) that an article was published.
- **author:** The person who wrote an article.
- **url:** The link to an article.
- **section:** The section of a publication (if applicable) in which an article appeared.
- **publication:** The name of the article’s publisher.

The following columns were kept, due to being directly relevant to our goal of identifying trends of stocks in different sectors:

- **date:** The date when an article was published.
- **title:** The title of an article.
- **article:** The text of an article. To save space and avoid unnecessary complexity, we truncated this to the first four sentences of each article.

The “title” and “article” columns were tokenized using the Natural Language Toolkit (NLTK), specifically the `word_tokenize()` and `sent_tokenize()` functions. After tokenizing and dropping unnecessary columns, the dataset was reduced to a size of 5.49 GB.

3.1.2. Stock Data

The stock prices used for the model were sourced from the Stock Market Dataset by Oleh Onyshchak on Kaggle. The dataset contains stock prices per day for all tickers that were traded on NASDAQ from 2016 to 2020, in CSV files sorted by the ticker symbol of each stock. The files in the original dataset were the folders “stocks” and “etfs”, along with “symbols_valid_meta.csv.”

The folder “etfs” represents exchange-traded funds, or a collection of various assets such as stocks, bonds, and commodities. Since ETFs provide this inherent diversification, they are generally considered less risky than stocks alone. However, since our fundamental goal is to minimize risk within the risky practice of stock trading, we do not want to pursue predictions with an inherently low risk, and so we discarded the “etfs” folder. Similarly, “symbols_valid_meta.csv” contains additional metadata for each ticker that is irrelevant to the actual stock prices, so it was also discarded.

The folder “stocks” contains 5,884 CSV files. To prevent the model from becoming too complex and failing to generalize to new data, we wanted to limit the number of stocks that were considered by the model. Therefore, we selected 10 of the most popular and relevant stocks from five different sectors, which are as follows:

1. Technology
 - a. AAPL (Apple Inc.)
 - b. NVDA (NVIDIA Corp.)
 - c. MSFT (Microsoft Corp.)
 - d. FB (Meta Platforms Inc. (Facebook))
 - e. GOOGL (Alphabet Inc. (Google))
 - f. AMZN (Amazon.com Inc.)
 - g. TSLA (Tesla Inc.)
 - h. AMD (Advanced Micro Devices)
 - i. ORCL (Oracle Corp.)
 - j. ADBE (Adobe)
2. Healthcare
 - a. JNJ (Johnson & Johnson)
 - b. PFE (Pfizer Inc.)
 - c. MRK (Merck & Co. Inc.)
 - d. ABBV (AbbVie Inc.)
 - e. LLY (Eli Lilly & Co.)
 - f. BMY (Bristol-Myers Squibb)
 - g. UNH (UnitedHealth Group Inc.)
 - h. GILD (Gilead Sciences Inc.)
 - i. ZBH (Zimmer Biomet Holdings)
 - j. CVS (CVS Health Corporation)
3. Financials
 - a. JPM (JPMorgan Chase & Co.)
 - b. BAC (Bank of America Corp.)
 - c. GS (Goldman Sachs Group Inc.)

- d. MS (Morgan Stanley)
 - e. WFC (Wells Fargo & Co.)
 - f. C (Citigroup Inc.)
 - g. AXP (American Express Co.)
 - h. SCHW (Charles Schwab Corp.)
 - i. USB (U.S. Bancorp)
 - j. BLK (BlackRock Inc.)
4. Energy
- a. XOM (ExxonMobil Corp.)
 - b. CVX (Chevron Corp.)
 - c. COP (ConocoPhillips)
 - d. EOG (EOG Resources Inc.)
 - e. ENPH (Enphase Energy Inc.)
 - f. FSLR (First Solar Inc.)
 - g. NEE (NextEra Energy Inc.)
 - h. DUK (Duke Energy Corp.)
 - i. SO (Southern Company)
 - j. VLO (Valero Energy Corp.)
5. Industrial
- a. BA (Boeing Co.)
 - b. CAT (Caterpillar Inc.)
 - c. GE (General Electric Co.)
 - d. DE (Deere & Co.)
 - e. UPS (United Parcel Service)
 - f. LMT (Lockheed Martin Corp.)
 - g. HON (Honeywell International)
 - h. RTN (RTX Corp. (Raytheon Tech))
 - i. UNP (Union Pacific Corp.)
 - j. MMM (3M Company)

This reduced the stock dataset from its original size of 2.75 GB to 43.9 MB.

We further processed this data by filtering the range of daily stock prices tracked by each CSV file from the range January 1, 2020 to April 1, 2020, which matches the range of news articles collected in the All the News 2.0 dataset. We additionally wanted to focus on the prices of stocks as a summary on a day-to-day basis; therefore, we dropped all columns except for “Date”, “Open”, and “Close”. When compiled into a single CSV, the size of the processed stock dataset is 2.91 MB.

3.1.3. Final News and Stocks Datasets

Here is the `pandas.head()` preview of the stocks DataFrame, which was saved to **stocks.csv**:

	stock	category	date	open	close
0	GOOGL	tech	2016-01-04	762.200012	759.440002
1	GOOGL	tech	2016-01-05	764.099976	761.530029
2	GOOGL	tech	2016-01-06	750.369995	759.330017
3	GOOGL	tech	2016-01-07	746.489990	741.000000
4	GOOGL	tech	2016-01-08	747.799988	730.909973

Here is the `pandas.head()` preview of the news article DataFrame, which was saved to **news.csv**:

	date	title	article
0	2016-12-09	['we', 'should', 'take', 'concerns', 'about', ...	[['this', 'post', 'is', 'part', 'of', 'polyarc...
1	2016-10-07	['colts', 'gm', 'ryan', 'grigson', 'says', 'an...	[['the', 'indianapolis', 'colts', 'made', 'and...
2	2018-01-26	['trump', 'denies', 'report', 'he', 'ordered', ...	[['davos', 'switzerland', 'reuters', 'us', 'pr...
3	2019-06-27	['france', 's', 'sarkozy', 'reveals', 'his', '...	[['paris', 'reuters', 'former', 'french', 'pre...
4	2016-01-27	['paris', 'hilton', 'woman', 'in', 'black', 'f...	[['paris', 'hilton', 'arrived', 'at', 'lax', '...

Note that while `news.csv` contains at least one row for every date in the range 2016-01-01 to 2020-04-01, `stocks.csv` is **sparse** and is not guaranteed to have stock data on a particular date in that range. However, we alleviate this issue through the way we collate our data.

3.2. Data Collation

Our news and stock data is collated into time-based context windows that are defined by the user. In our preprocessing code, the function `get_news_windows` takes two parameters:

- **window_size** determines the length of one window, in days.
- **num_windows** determines how many windows will be collected.

From the total range of our data (January 1, 2016 to April 1, 2020), a random starting date is selected. Data from each window is collected starting from this date and moving forward. For example, if `window_size = 2`, `num_windows = 3`, and the starting date is January 5, 2016, these context windows are collected:

- [2016-01-05, 2016-01-07)
- [2016-01-06, 2016-01-08)
- [2016-01-07, 2016-01-09)

Each context window contains two elements:

- **headlines**: A list of tokens from all article titles within that window.
- **delta**: A label that indicates the average movement of all of our selected stocks. It can take one of three possible values:
 - “GAIN”: The stock prices increase on average.
 - “LOSS”: The stock prices decrease on average.
 - “FLAT”: The stock prices remain the same on average.

The context windows are saved into a CSV file named “`final.csv`”, which is of arbitrary size depending on the window length parameters chosen by the user.

3.4. Model Architecture

We experiment with two feature extraction methods: TF-IDF, which captures word frequency and is fast to train on, and GloVe embeddings, which capture semantic meaning of words by averaging their 200-dimensional vectors across tokens. These are fed into a multi-layer perceptron (MLP), implemented using `sklearn`'s `MLPClassifier`. The architecture consists of two hidden layers (either 50 or 100 neurons each), ReLU activations, Adam optimizer, and hyperparameters like learning rate (0.0001 or 0.01) and L2 regularization ($\alpha = 0.0001$ or 0.01) tuned via 5-fold cross-validation. Although `sklearn` does not expose batch size or epochs directly, it uses full-batch training with early stopping based on tolerance.

We selected the MLP because it balances expressive power and training simplicity for vectorized data. Evaluation is done using macro F1 score, which is appropriate due to label imbalance (e.g., FLAT being underrepresented), and accuracy. The methods chosen are reasonable because they represent both a baseline (TF-IDF + MLP) and a semantic-aware approach (GloVe + MLP), allowing us to compare frequency-based and meaning-based representations on their ability to model financial outcomes from news.

4. Experiments

We trained a classification model to predict short-term stock market movement—categorized as GAIN, LOSS, or FLAT—based on tokenized news headlines. The dataset was generated using a sliding time window over news and stock data, and split into training and testing sets using an 80/20 split with a fixed random seed of 42 for reproducibility. Within the training set, we performed 5-fold cross-validation to fine-tune hyperparameters. The model is a multi-layer perceptron (MLP) with two hidden layers (either 50 or 100 neurons), ReLU activation, and trained using the Adam optimizer with learning rates of 0.0001 or 0.01 and L2 regularization ($\alpha = 0.0001$ or 0.01). Input features were derived from either TF-IDF vectors or GloVe word embeddings. Evaluation was conducted using both accuracy and macro F1 score, the latter being more appropriate due to mild class imbalance.

Our model achieved 80–100% accuracy, which is high but expected, as many news events like major earnings announcements, geopolitical conflicts, or economic indicators have consistent and repeatable effects on market behavior. However, not all news behaves predictably—ambiguous or volatile headlines like political statements or rapidly evolving crises can have inconsistent market impacts, leading to misclassifications. Furthermore, overlapping or repeated headlines across time windows may cause the model to overfit, memorizing surface patterns rather than learning generalized market behavior.

From this experiment, we learned that headline semantics alone can be highly predictive, but model performance is sensitive to data leakage, temporal overlap, and labeling granularity. It suggests the need for more sophisticated temporal models (e.g., LSTMs) or richer input features (e.g., sentiment scores, stock-specific embeddings) in future work.

Works Cited

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